

Characterizing Environmental Severity and Atmospheric Corrosivity Categorization in Alaska

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ABSTRACT

Atmospheric corrosion continues to be a significant monetary problem. Particularly in colder climates, such as Alaska, the complexity of atmospheric corrosion increases and additional factors make it difficult to estimate the corrosive phenomena. Despite the topic's significance, there exists a noticeable knowledge gap regarding the corrosivity of metals in cold regions in general. This lack of data stands in stark contrast to Alaska's growing importance on the global stage, particularly its potential as a pivotal point for cargo, strategic military location, refueling hub, and location for aerospace applications. Multi-angle corrosion test racks were deployed at four test sites across Alaska, each distinct in their environment, equipped with weather sensors and chloride candles. The principal weather parameters recorded were the ambient air temperature, relative humidity, time of wetness at exposure angles of 0°, 30°, and 45°, total rainfall, and aerosol chloride and sulfate deposition. Once coupon exposure was complete, coupon mass loss was measured and the corrosion rates were calculated as a function of the surface exposure angle. Comprehensive analyses, including weather correlation, mass loss analysis, SEM/EDXA, and optical profilometry, were used to determine corrosion rates and classify the corrosivity of the four sites against well-recognized standards.

Key words: Alaska, Arctic Regions, Atmospheric Corrosion, Dose Response

INTRODUCTION

Background

Corrosion is a ubiquitous and complex process in which metals and alloys deteriorate under environmental influences. It represents the inherent tendency of metals to revert to their more stable oxide, hydroxide, or sulfide forms. While corrosion is a naturally occurring phenomenon, its implications span across both economic and functional dimensions. Economically, corrosion has been estimated to

cost billions of USD every year in terms of material losses and the subsequent replacement of infrastructure, machinery, and other assets. Functionally, the integrity and longevity of structures, tools, and equipment can be compromised, leading to unexpected failures that could result in safety hazards. Moreover, the environmental fallout of corroded structures can introduce toxins and pollutants into ecosystems. To curtail these consequences, understanding the fundamental nature, mechanisms, and prevention techniques of corrosion becomes paramount.¹

Atmospheric corrosion is particularly prevalent and concerning to industries that expose metals to the environment. At its core, atmospheric corrosion is an intricate interplay between the metal, the surrounding environment, and the ever-present moisture. Factors such as humidity, temperature, air pollutant concentration, and time of wetness (TOW) play significant roles in dictating the rate and nature of corrosion. The ubiquity of atmospheric conditions means that understanding this form of corrosion and its mechanisms is paramount for ensuring the longevity and safety of exposed infrastructures.² Previous research has made significant strides in understanding atmospheric corrosion. The International Atmospheric Exposure Program (ISO⁽¹⁾ CORRAG) collected data over 12 years at 53 test sites across Europe, North America, and Asia. It has the largest collection of atmospheric corrosion and environmental data ever published in a single source and led to the creation of ISO⁽¹⁾ 9223:1992 (updated to ISO⁽¹⁾ 9223:2012), a key standard for classifying the corrosivity of different atmospheres.³

The dynamics of atmospheric corrosion in colder regions introduce additional challenges. Lower temperatures just above freezing can lead to prolonged TOW due to dew or frost, extending the period metals are exposed to moisture, a key factor for corrosion. Further, salt-induced corrosion, commonly seen in coastal marine regions, is replicated in colder inland areas due to the use of de-icing salts. Snow and ice can also trap pollutants close to the metal surface, allowing for corrosive conditions.^{4&5} These factors combined mean that the traditional understandings of corrosion rates may not directly apply in colder climates.⁶ Research conducted in the colder climate surrounding Canada analyzes the dose-response function, found in standard ISO⁽¹⁾ 9223:2012 used to estimate corrosion rates of metals, and compared it to experimentally obtained results.^{7&8}

Approach and Objectives

Multi-angle corrosion test racks equipped with weather sensors and chloride candles were deployed at four test sites to expose metal alloy coupons, each test site representing the distinct environments across Alaska. The choice of metal samples is based on materials that are widely used in land, sea and aerospace transportation and infrastructure industries. Weather parameters recorded include the ambient air temperature, relative humidity, time of wetness at exposure angles of 0°, 30°, and 45°, rainfall, and aerosol chloride and sulfate deposition. Upon completion of coupon exposure, mass loss due to corrosion was measured and corrosion rates were calculated as a function of the surface exposure angle. The corrosion rates for carbon steel were then correlated with the recorded weather parameters and compared to standard methods of classifying atmosphere corrosivity. Objectives of this study include:

- To assemble and deploy multi-angle corrosion test racks with auxiliary weather stations at four test sites across Alaska, USA.
- To monitor and measure the ambient air temperature, relative humidity, time of wetness at exposure angles of 0°, 30°, and 45°, rainfall, and aerosol chloride and sulfate deposition at the test sites across 6-month, 12-month, and 24-month exposure periods. Only the 12-month exposure period results are discussed in this paper.
- To expose five different metal alloys to atmospheric conditions in Alaska at 0°, 30°, and 45° exposure angles. Only 1008 carbon steel (UNS⁽²⁾ G10080) is discussed in this paper.

⁽¹⁾ International Organization for standardization (ISO), 1241 Vernier, Geneva, Switzerland

⁽²⁾ Unified Numbering System for Metals and Alloys (UNS). UNS numbers are listed in Metals & Alloys in the Unified Numbering System, 10th ed. (Warrendale, PA: SAE International and West Conshohocken, PA: ASTM International, 2004).

- To correlate the corrosion rate of 1008 carbon steel to exposure conditions and weather parameters to determine the corrosivity category and compare to recognized standards.

EXPERIMENTAL PROCEDURE

Test Rack and Weather Station

The design and methodology of atmospheric corrosion test racks has been guided by several standards over the years, prominently, ASTM⁽³⁾ G50: “Standard Practice for Conducting Atmospheric Corrosion Tests on Metals”⁹ and ISO⁽¹⁾ 8565: “Metals and alloys – Atmospheric corrosion testing – General requirements.”¹⁰ The bench-top test rack used in this study has been developed in accordance with ASTM⁽³⁾ standards for atmospheric corrosion research and has been designed with a primary objective of supporting snow loads up to 1000lbs. expected over multiple winters in Alaska (Figure 1). This design, built with adjustable aluminum alloy railings, allows for adaptability across varied ground conditions, directional alignment, and angle of exposure for coupons, central to the experiment.

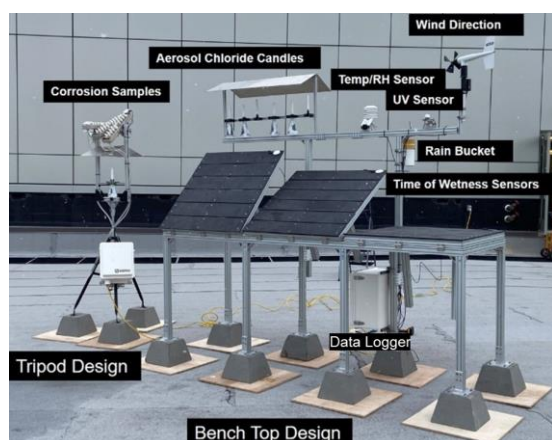


Figure 1: Multi-Angle Corrosion Test Rack Bench-Top Design with Auxiliary Weather Station



Figure 2: Map of Alaska with Test Site Locations

Additionally, the bench-top design can support a full weather monitoring system rather than relying on the nearest meteorological observatories to capture weather data, as shown in Figure 1. This removes the reliance on observatories which could be located miles away from the test site, increasing the accuracy of weather parameters related to coupon exposure.

Test Sites and Exposure Periods

Four strategic locations were selected as coupon exposure test sites, with test site selection and characterization influenced by ASTM⁽³⁾ G92 “Standard Practice for Characterization of Atmospheric Test Sites.”¹¹ These locations, spread around Alaska, were chosen to obtain a set of meteorological profiles that represent the diverse atmosphere and environment of Alaska along with their significance at the national and state level, summarized in Table 1. The locations of the test sites can be referenced in Figure 2, a map of the state of Alaska.

Pacific Spaceport Complex – Alaska (PSCA), Kodiak, AK: This site offers an aggressive marine environment and caters to both the fishing and aerospace sectors. Unique in its standing, the spaceport is the only high-latitude operational spaceport in the U.S. With an annual rainfall of 81 inches (~2000 mm), quadruple the national average, this location offers insight into prolonged time of wetness at lower temperatures. This is the only test site located on the ground and sits within a fenced area 165 feet (50 meters) above sea level near a lagoon.

⁽³⁾ American Society for Testing and Materials (ASTM) Headquarters. 100 Barr Harbor Drive P.O. Box C700 West Conshohocken, PA 19428

University of Alaska Fairbanks (UAF), Fairbanks, AK: Representing a far-inland urban environment, this location has a distinct meteorological profile with fewer aerosols compared to coastal sites and experiences colder ambient air temperatures. This test site is located on the roof of the Joseph E. Usibelli Engineering Building at UAF, a six-story building. The height of the test site ranges from 60–72 feet (18–22 meters) above ground level.

Port of Alaska (PAA), Anchorage, AK: Chosen for its moderate marine environment, this location is strategically positioned on the coastline serving the fuel, freight, and cruise ship sectors. Meteorologically, it offers conditions similar to the UAA test site with anticipated variations in aerosol content. This test site is located on the roof of the Port of Alaska Administration Building, a three-story building. The height of the test site ranges from 30–36 feet (9–11 meters) from the building's foundation which sits another 10–13 feet (3–4 meters) above the water line for a total of 40–50 feet (12–15 meters) above the sea level.

University of Alaska Anchorage (UAA), Anchorage, AK: Located on the fourth level (roof) of the Engineering Parking Garage, this location represents an inland mild-marine environment. Located in the heart of Alaska's largest city, it is approximately 5 miles from one of the world's top cargo airports and less than 4 miles from the coastline. The elevation above ground is estimated to be around 30–45 feet (9–13 meters).

Table 1: Test Site Coordinates, Distance from Salt Water, Elevation, and Environment

Location	Coordinates	Distance to Nearest Body of Salt Water	Elevation AMSL	Environment
Pacific Spaceport Complex – Alaska	57.453°N, 152.346°W	~600 feet (183 meters) from Pacific Ocean	0 feet (0 meters)	Aggressive Marine Environment
University of Alaska Fairbanks	64.855°N, 147.817°W	~250 miles (400 km) from Pacific Ocean ~350 miles (560 km) from Arctic Ocean	60–72 feet (18–22 meters)	Far-inland Urban Environment
Port of Alaska	61.239°N, 149.888°W	~30 feet (9 meters) from Pacific Ocean	30–36 feet (18–22 meters)	Moderate Marine Environment
University of Alaska Anchorage	61.192°N, 149.822°W	~4 miles (6.4 km) from Pacific Ocean	30–45 feet (9–13 meters)	Inland Mild-Marine Environment

Material Selection

1008 carbon steel (UNS⁽²⁾ G10080) coupon preparation consisted of sandblasting the surfaces with 80 grit aluminum oxide media, cleaning with compressed air, pin stamping for coupon identification, degreasing and cleaning with acetone, and preliminary weighing. The 1 inch x 2 inch, 0.125 inch thick (25.4 mm x 50.8 mm, 3.175 mm thick) coupons were each mounted on an aluminum plate using low contact, non-conductive hardware as shown in Figure 3. This method of mounting avoids alteration in the coupon thus avoiding variability in corrosion. Two stainless steel dividers isolate the carbon steel coupons to prevent cross-contamination of corrosion products. Each test site contained four coupons for each exposure period at each exposure angle. This study will present and discuss the analysis of the corrosion of 1008 carbon steel while the additional metal alloys exposed will be studied in future works. The chemical composition of 1008 carbon steel can be found in Table 2.

Table 2: Chemical Composition of 1008 Carbon Steel (UNS⁽²⁾ 10080)

Iron, Fe	Manganese, Mn	Carbon, C	Sulfur, S	Phosphorous, P
99.30–99.70	0.30–0.50	0.10	0.05	0.04

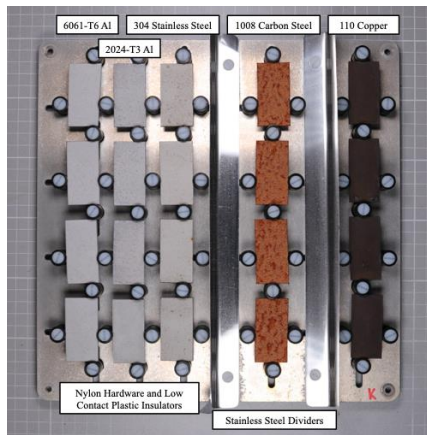


Figure 3: Metal Alloys Mounted with Hardware and Dividers

Determining Corrosion Rates

Coupons, post-exposure, were chemically cleaned according to ASTM⁽³⁾ G1 “Standard Practice for Preparing, Cleaning, and Evaluating Corrosion Test Specimens.”¹² Each coupon mass was taken between each cleaning to ensure corrosion product was being removed while a stable mass was being approached. For the cleaning of 1008 carbon steel, the chemical cleaning solution consisted of 500 mL of hydrochloric acid (HCl), 3.5 g of hexamethylenetetramine, and de-ionized water to make 1000 mL. A pristine reference coupon was included in each respective batch of cleaning to verify the cleaning procedure was not significantly attacking any base metal and impacting its final mass measurement. After the final cleaning, the coupons were weighed and the corrosion rate, in grams per meter squared per day (gmd), was calculated according to Equation 1 where “M” represents the coupon mass in grams, “D” represents the exposure period in days, and “SA” represents the total surface area of the coupon in square meters.

$$\text{Corrosion Rate} = \frac{M_{\text{initial}} - M_{\text{final}}}{SA \cdot D} \quad (1)$$

SEM, EDXA, and Surface Morphology

For each exposure angle at each test site, a single carbon steel coupon, not subject to the cleaning procedure, was saved for further analysis. At a center location, similar on each coupon, scanning electron microscope (SEM) imagery and energy dispersive x-ray analysis (EDXA) of the corrosion product was conducted. Each were scanned at 100X magnification and both the weight % and atomic % of the elements were reported. Optical profilometry was utilized to obtain the surface morphology of the damage caused to each carbon steel coupon. The axial chromatism technique was followed.¹³ Before optical profilometry analysis, all corrosion product was removed from the coupon, cleaned according to ASTM⁽³⁾ G1. The scanned center of each coupon is a 10 mm x 10 mm area at 5 μm increments in both the x and y directions. Post-processing techniques involved the creation of contour plots and undergoing a surface roughness analysis based on ISO⁽¹⁾ 25178 “Geometrical product specifications (GPS) – Surface texture: Areal” providing the guidelines and definitions for the measurement and characterization of the three-dimensional surface texture.¹⁴ A MATLAB[†] script was then used to quantify the material defect percentages within the contour plots.

RESULTS AND DISCUSSION

Weather Data

Figures 4-6 visually present weather data from the four test sites for the first set of 12 months, with the ambient air temperature and relative humidity being mean adjusted and the time of wetness representing the percent of time the coupons are considered wet over the total exposure period. In early

[†] Trade name

field exposure programs, the conventional approach predominantly involved using time of wetness as the primary explanatory factor of corrosion, overlooking the effects of air temperature and relative humidity.¹⁵ It has been found in previous studies that a positive correlation between the corrosion rate and the air temperature exists until approximately 10°C, varying with the material, beyond which a negative correlation occurs. It is also important to note that when colder temperatures prevail, and certain aerosol particles are present, the freezing point of a liquid layer on a metal's surface is normally reduced to far below freezing and significant corrosion rates can be experienced down to -5°C.¹⁶

Figure 4 presents the ambient air temperature localized at the exact location of the test site, each following trends throughout the distinct seasons experienced in Alaska. It was found that the PSCA test site experienced the warmest average ambient air temperatures, remaining predominately above 0°C, throughout the entirety of the 12-month exposure period. The UAF test site contrasts this with intense shifts in temperature with equal time above and below the point of freezing throughout the seasons. This test site experienced extremes down to -35°C in the winter and up to 28°C in the summer with the average ambient air temperature existing at just around 0°C. Such results promote further research into aligning the exposure periods with the seasons. This would encapsulate either solely the extremes of warmer or colder months, fostering insight into the seasonal impact on atmospheric corrosion. Being approximately 4 miles apart, the PAA and UAA test sites exhibited comparable ambient air temperature profiles, both establishing an average around 4°C. The combined temperature readings provide a clear illustration of the relative fridity of the climates in Alaska in comparison to other locations typically studied for atmospheric corrosion, underscoring Alaska's distinctive colder, Arctic, and sub-Arctic conditions. It is also to be noted that though ambient temperatures remained low during the winter season, the metallic surfaces of the coupons may attain significantly higher temperatures due to solar radiation, prompting further research and possibly inclusion of this additional parameter.

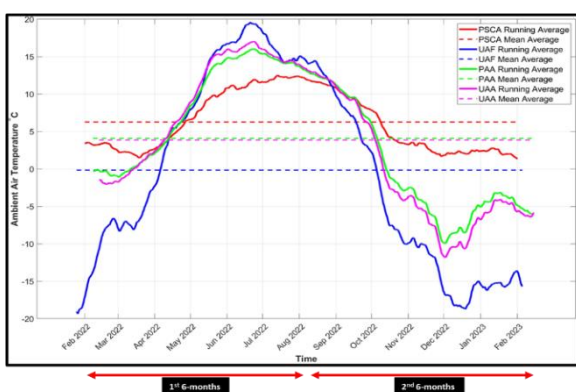


Figure 4: Ambient Air Temperature at each of the Four Test Sites over the First 12-Month Exposure Period

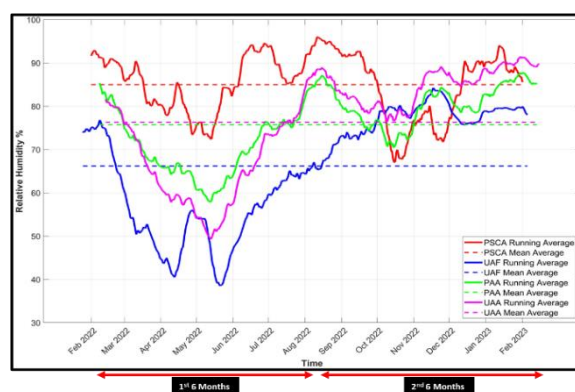


Figure 5: Relative Humidity at each of the Four Test Sites over the First 12-Month Exposure Period

Relative humidity is an important factor in analyzing the corrosivity of a particular atmosphere, especially when coupled with the presence of hygroscopic contaminants, fostering an environment for corrosion on metal surfaces. The combined factors create a dynamic “critical relative humidity”. This is a threshold at which condensation occurs, quantified as a period known as the “time of wetness”. It can be seen in Figure 5 that the relative humidity the PSCA test site experiences is comparatively stable, trending above 70% throughout the 12-month exposure period. This can be attributed to its stabilizing effect on air temperature due to its proximity to a large body of water, the Pacific Ocean. In comparison, the UAF test site experienced the lowest average relative humidity with a spike in the month of April due to an unexpected deluge of rainfall over the course of 2-3 weeks. The PAA and UAA test sites exhibited comparable relative humidity profiles, though not completely identical. For these two test sites, the relative humidity follows an inverse trend to the air temperature, the UAA test site being less humid and warmer in the summer and more humid and cooler in the winter.

As mentioned previously, the “critical relative humidity” quantifies a period known as the “time of wetness”. This further refers to the interval when a corrosive liquid film is present on a surface. Using leaf time of wetness sensors deployed at exposure angles of 0°, 30°, and 45° from the horizontal, this method can determine when the surface of the metal coupon is expected to be moist and susceptible to corrosion. This extends beyond the conventional method of defining this period as when the ambient temperature exceeds 0°C while the relative humidity reaches 80% or higher. Figure 6 presents the total time of wetness as a percentage at each test site for the first 12-month exposure period, most notably differentiating between the exposure angles. It should also be noted that the sensors deployed at the UAA test site malfunctioned. This data was deemed inaccurate and brings up the condition of ice buildup and snow interfering with weather data collection.

As demonstrated across all test sites, the time of wetness was found to inversely correlate with the exposure angle. The metal coupons deployed at a 0° exposure angle resulted with a higher time of wetness than those at a 30° and 45° angle of exposure. This intuitively reflects gravity’s influence on water droplets where water accumulation would diminish as the surface angle increases from the horizontal. The ISO⁽¹⁾ 9223 standard indicates that time of wetness estimations can be closely inferred from existing temperature and relative humidity data.¹⁸ With this, the PSCA test site registered the highest time of wetness when compared to the other three test sites, corresponding with its highest average ambient temperature and highest average relative humidity. In opposite way, the UAF test site recorded the lowest time of wetness. There is a positive correlation between the time of wetness and relative humidity at each test site while the correlation of temperature against other parameters studied do not adhere to anticipated patterns. This may be attributed to Alaska’s subfreezing temperatures.

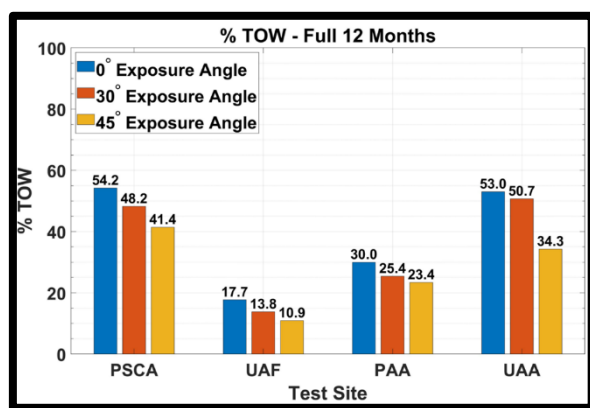


Figure 6: Percent Time of Wetness at each of the Four Test Sites over the First 12-Month Exposure Period at each Angle of Exposure

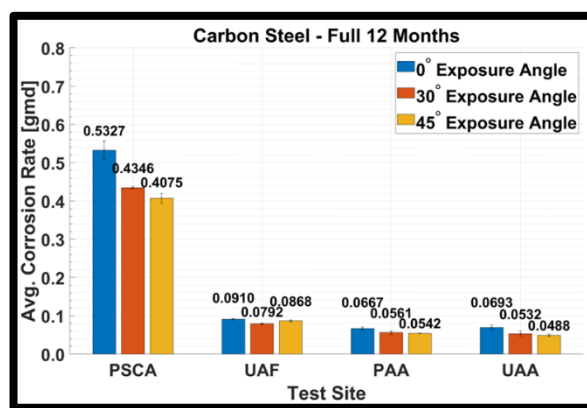


Figure 7: Average Corrosion Rates of 1008 Carbon Steel at each of the Four Test Sites over the First 12-Month Exposure Period at each Angle of Exposure

Known to be the two leading contaminants that impact corrosion, chloride ions and sulfates can sustain presence on metal surfaces due to the hygroscopic nature of corrosion products and aerosol deposits. Table 3 presents the aerosol chloride and sulfate deposition data gathered according to ASTM⁽³⁾ G140 “Standard Test Method for Determining Atmospheric Chloride Deposition Rate by Wet Candle Method” and analyzed using ion chromatography.¹⁷ Predicted to have the highest chloride and sulfate deposition rates due to its proximity to the ocean, the PSCA test site resulted with four to nine times that of the UAF, PAA, and UAA test sites. Also located near a body of salt water, the PAA test site recorded lower chlorides than expected, encouraging further study into the wind patterns and surf zone of the location. Main sources of chloride ions are the ocean and de-icing salts used on roads and the main sources of sulfates are the burning of coal, wood, and oil. This aligns with the increased sulfate deposition rates recorded in the colder months within the 6-month exposure period weather data set.

Highlighting its multifaceted effects, Leygraf et. al. offers a comprehensive account of the impact that rainfall or precipitation has on atmospheric corrosion¹⁶. This is due to its ability to both wash away

accumulated corrosive agents and introduce corrosive ions through wet deposition on the surface of a material. The overall consequence being contingent upon specific environmental conditions and the nature of the material. Though, an additional layer of complexity can be attributed to this parameter as the effectiveness of the cleansing action of rain on dry deposits is not solely reliant on precipitation patterns but also reliant on the characteristics of the corrosion products. Over time, these products develop more complex surface morphologies with enhanced capacity to hold onto deposited materials¹⁶. It is noted that the specific rain gauge used in this study encountered operational challenges in subzero conditions. Table 3 provides a summary of the weather data obtained.

Table 3: Summary of Weather Data at each Test Site over the First 12-Month Exposure Period

Location	Average Air Temperature [°C]	Average Relative Humidity [%]	Time of Wetness [%]			Chloride Deposition [mg/(m ² ×day)]	Sulfate Deposition [mg/(m ² ×day)]
			0°	30°	45°		
PSCA	6.26	85	54.2	48.2	41.4	95.4	20.3
UAF	-0.07	66.8	17.7	13.8	10.9	18.5	6
Port of Alaska	8.86	77.1	30	25.4	23.4	22.7	5
UAA	9.64	76.3	—	—	—	17.7	3.8

Corrosion Rates

Figure 7 depicts the corrosion rates for 1008 carbon steel at each of the four test sites at each angle of exposure over the first 12-month exposure period. Following coupon exposure, three out of the four coupons were cleaned of corrosion product and weighed, having their mass loss averaged to determine the corrosion rate. The fourth coupon was reserved for SEM imagery and EDX analysis. It can be observed that the carbon steel coupons exposed at the PSCA test site exhibited corrosion rates four times greater than the carbon steel coupons exposed at the UAF, PAA, and UAA test sites. This contrast can be attributed to the weather, where the PSCA test site recorded four times the chloride deposition rate as well as the coupons having spent 18% more time wet than the others. When comparing the first 12-month exposure period of each of the test sites, the PSCA test site also recorded the highest average air temperature, average relative humidity, total rainfall, and sulfate deposition rates. A visual of the corroded coupons from the first 12-month exposure period shows that the coupons exposed at the PSCA test site accumulated more corrosion product, covering the entirety of their surfaces, than those exposed at the UAA test site (Figure 8).

Upon inspection, a distinct correlation was found between the exposure angle and the corrosion rate. Coupons exposed at 0° showed the highest corrosion rates, followed by those exposed at 30°, with the lowest corrosion rates found on coupons exposed at 45°. Although the coupons exposed at the UAF, PAA, and UAA test sites exhibited less pronounced trends in corrosion and experienced corrosion rates four times less than those of coupons exposed at the PSCA test site, the coupons deployed at a 0° exposure angle consistently resulted with higher corrosion rates than those of the coupons deployed at the 30° and 45° angle of exposure. This result further strengthens the argument that the angle at which the surface of a metal is exposed to the atmosphere affects its experience with corrosion.

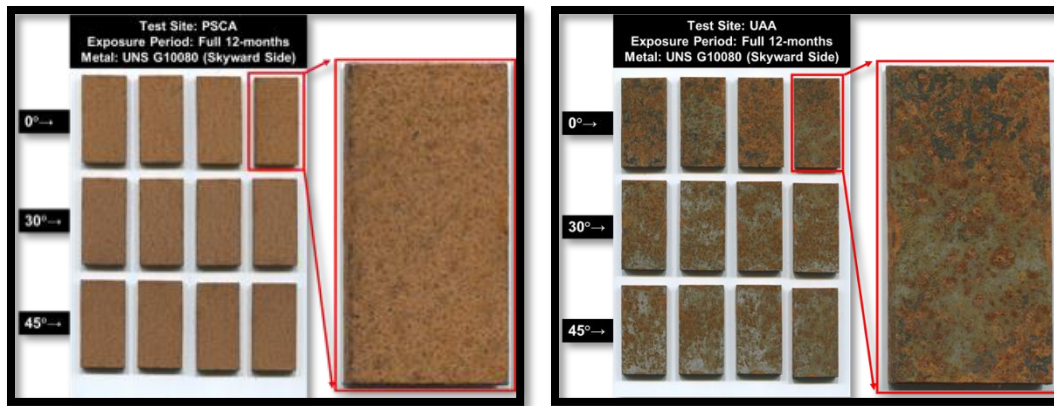


Figure 8: Scan of Skyward Side of 1008 Carbon Steel Coupons Exposed over the First 12-Month Exposure Period at the PSCA (left) and the UAA (right) Test Sites

Corrosion Rate Correlation

Time of wetness and chloride deposition rates were chosen for further analysis and correlation against the corrosion rates as they are known primary factors that influence the corrosion process of metals. Figure 9 presents the correlation between the corrosion rates of 1008 carbon steel and the time of wetness at each test site. Figure 10 presents the correlation between the corrosion rates of 1008 carbon steel and the chloride deposition rates. In this figure, the corrosion rates were averaged across the three exposure angles to align with the singular chloride deposition rate recorded at each site for each exposure period. Both figures contain the first and second 6-month and first 12-month exposure periods. Fit lines were added in the presented analysis where the red lines represent the correlations including all four test sites while the blue lines represent the correlations excluding the PSCA test site. Including all four test sites demonstrates strong positive correlations suggesting that increases in time of wetness and chloride deposition rates induce higher corrosion rates. Conversely, excluding PSCA demonstrates negligible or moderately negative correlations. This contrast can be attributed to the significantly lower time of wetness and chloride deposition rates experienced by the metal coupons at the UAF, PAA, and UAA test sites leading to lower corrosion rates. Additionally, lower temperatures may also be a contributing factor to the minimal and negative correlations observed by these three test sites.

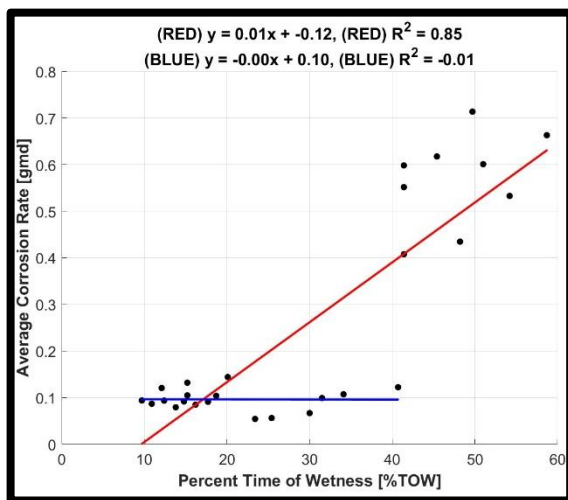


Figure 9: Correlation between % Time of Wetness and First Year Corrosion Rates of 1008 Carbon Steel

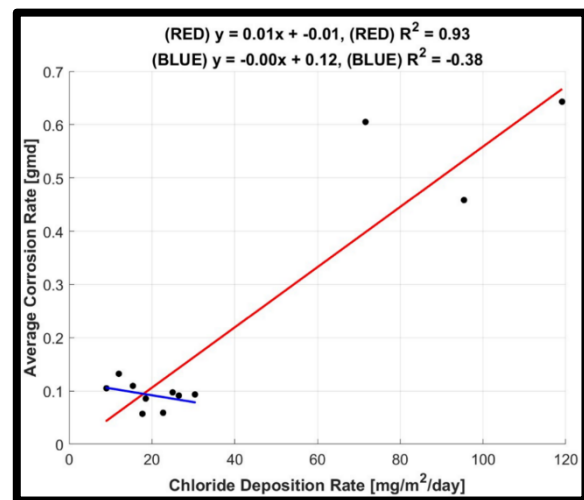


Figure 10: Correlation between Chloride Deposition Rates and First Year Corrosion Rates of 1008 Carbon Steel

Corrosivity Category and Corrosion Estimation

The ISO⁽¹⁾ 9223 standard states that the corrosivity of an atmospheric environment shall be classified through determination of the corrosivity category either based on experimental exposure of specimens or by estimation based on weather parameters and the use of a dose-response function.¹⁸ For carbon steel, the level of uncertainty for determining the corrosivity category of an atmosphere through the method of exposure of specimens is $\pm 2\%$ while the level of uncertainty through the method of estimation with the dose-response function is -33% to $+50\%$. The investigation conducted presented the opportunity to assess the correlation between the two methods of determination. As provided in the ISO⁽¹⁾ 9223 standard, the corrosivity based on first year corrosion rates can be determined when calculated using mass loss. Table 4 categorizes the corrosivity of an atmosphere with C1 being very low and CX being extreme.

Table 4: Corrosion Rates and Corrosivity Category for the First Year of Exposure of Carbon Steel¹⁸

Corrosion Rate, r_{corr} [g/(m ² ×year)]	Corrosivity Category	Corrosivity
≤ 10	C1	Very low
$10 < r_{\text{corr}} \leq 200$	C2	Low
$200 < r_{\text{corr}} \leq 400$	C3	Medium
$400 < r_{\text{corr}} \leq 650$	C4	High
$650 < r_{\text{corr}} \leq 1500$	C5	Very high
$1500 < r_{\text{corr}} \leq 5500$	CX	Extreme

Table 5 presents the corrosion rates for 1008 carbon steel at each of the four test sites over the first 12-month exposure period for each angle of exposure, expressed in g/m²/year, along with the corresponding corrosivity category. In accordance with the ISO⁽¹⁾ 9223 standard, the results show that each test site classified within the C2, low corrosivity category. However, it was found that the corrosion rate experienced at the PSCA test site approaches the upper boundary of the C2 category. The dose-response function within the ISO⁽¹⁾ 9223 standard serves as a valuable tool in the determination of the corrosivity category of an atmosphere when the exposure of specimens is deemed impractical. This tool provides an estimated corrosivity category and range of corrosion rate based on a set of weather parameters, including ambient air temperature, average relative humidity, and chloride and sulfate deposition rates. The standard also specifies four distinct dose-response functions tailored to carbon steel, zinc, copper, and aluminum. Equation 2 outlines the method for calculating the estimated first-year corrosion rate for carbon steel where r_{corr} represents the first-year corrosion rate in micrometers per year, T represents the annual average temperature in degrees Celsius, RH represents the annual average relative humidity expressed as a percentage, P_d represents the annual average SO₂ deposition in milligrams per square meter per day, and S_d represents the annual average Cl⁻ deposition in milligrams per square meter per day.

$$r_{\text{corr}} = 1.77 \times P_d^{0.52} \times e^{0.02 \times RH + f_{\text{St}}} + 0.102 \times S_d^{0.62} \times e^{0.033 \times RH + 0.040 \times T} \quad (2)$$

$$f_{\text{St}} = 0.150 \times (T - 10) \text{ when } T \leq 10^\circ\text{C}; \text{ otherwise } -0.054 \times (T - 10)$$

Using Equation 2, the estimated corrosion rate for 1008 carbon steel was calculated with the average annual weather parameters for each test site at the 30° exposure angle. The estimated first-year corrosion rate, based solely on the four weather parameters, is compared to the first-year corrosion rate determined by mass loss, in Table 6. From the table, the dose-response function estimates the first-year corrosion rates to be two to three times greater than the corrosion rates determined through coupon exposure for the PSCA and UAF test sites and six to seven times greater for the PAA and UAA test sites. Despite the stark differences between the corrosion rates of the two methods, the corrosivity category for

the UAF, PAA, and UAA test sites remain in the C2, low, corrosivity category while the PSCA test site increases to the C4, high, category.

Table 5: Corrosivity Category for each Test Site Determined by Coupon Exposure

Location	1008 Carbon Steel UNS G10080		
	Exposure Angle	Corrosion Rate [g/(m ² ×year)]	Corrosivity Category
Pacific Spaceport Complex – Alaska	0°	194.4	C2
	30°	158.6	
	45°	148.7	
University of Alaska Fairbanks	0°	33.2	
	30°	28.9	
	45°	31.7	
Port of Alaska	0°	24.3	
	30°	20.5	
	45°	19.8	
University of Alaska Anchorage	0°	25.3	
	30°	19.4	
	45°	17.8	

Table 6: Corrosivity Category of 1008 Carbon Steel determined by Coupon Exposure and the Dose-Response Function

Location	1008 Carbon Steel				
	Exposure Angle	Corrosion Rate [g/(m ² ×year)] (Coupon Exposure)	Corrosivity Category	Corrosion Rate [g/(m ² ×year)] (Dose-Response Function)	Corrosivity Category
Pacific Spaceport Complex – Alaska	30°	158.6	C2	494.6	C4
University of Alaska Fairbanks	30°	28.9		73.8	C2
Port of Alaska	30°	20.5		146	
University of Alaska Anchorage	30°	19.4		119.9	

SEM, EDXA, and Surface Morphology Data

One carbon steel coupon from each test site for each exposure period at each angle of exposure was analyzed at one center location on the skyward side to examine the corrosion behavior at the surface. The surface pattern found on the pristine sandblasted coupons was observed on coupons from the UAF, PAA, and UAA test sites, however not discernible on the coupons exposed at the PSCA test site (Figure 11). This is due to the surfaces being entirely encased in an iron oxide layer. For this reason, the PSCA coupons also posted the highest oxygen percentages as can be seen in Table 7. This can also be visually

confirmed through the high-resolution scans previously presented in Figure 8. The summarized weight percentages of the 1008 carbon steel coupons at each test site for the first 12-month exposure period at each exposure angle in Table 7 indicate that coupons from the UAF, PAA, and UAA test sites, along with a pristine sandblasted coupon, contained trace levels of aluminum and silicon, while conversely, a pristine non-sandblasted coupon exhibited no detectable amounts. This can be traced back to the sandblasting medium used. Due to the full surfaces of the coupons exposed at the PSCA test site being covered in an iron oxide layer, no detectable amounts of aluminum or silicon were found. This raises the potential need to revise the procedure for surface preparation to prevent contamination of foreign elements before coupon exposure.

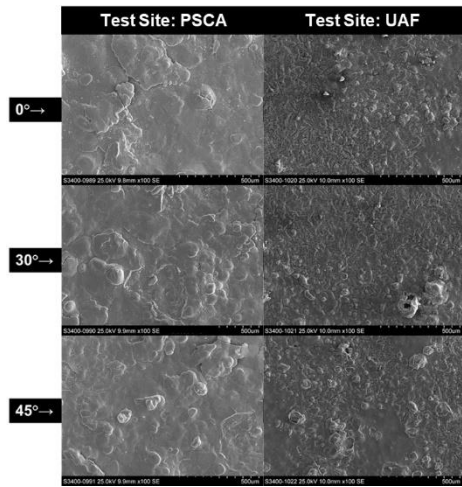


Figure 11: SEM Scans of Corroded 1008 Carbon Steel after the First 12-Month Exposure Period — PSCA (left) and UAF (right)

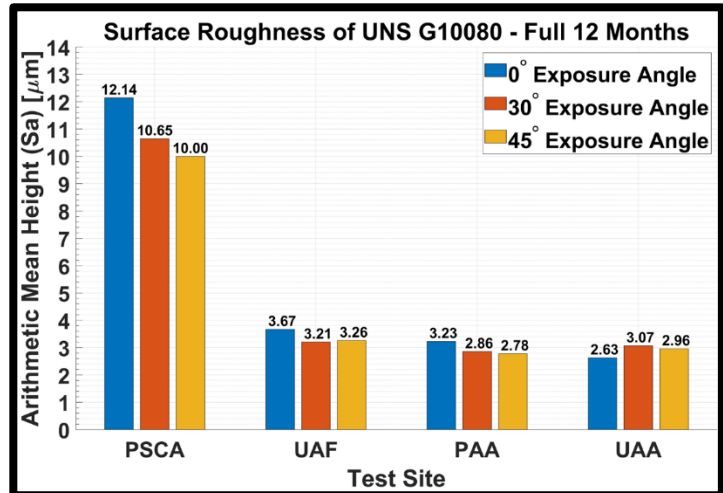


Figure 12: Surface Roughness of 1008 Carbon Steel at each of the Four Test Site over the First 12-Month Exposure Period at each Angle of Exposure Post-Cleaning

Table 7: EDXA — Weight Percent Summary of 1008 Carbon Steel Coupons from each Test Site for the First 12-Month Exposure Period

Test Site	Angle	EDXA — Elements by Weight Percent							
		C	O	Al	Si	Cl	Ti	Mn	Fe
Pristine Non-Sandblasted	n/a	9.79	—	—	—	—	—	0.66	89.55
Pristine Sandblasted	n/a	15.41	8.69	4.8	0.2	—	0.12	0.6	70.18
PSCA 1st 12 Month Exposure Period	0°	—	37.45	—	0.18	0.11	—	0.37	61.89
	30°	—	37.34	—	—	0.12	—	0.39	62.16
	45°	—	36.73	—	—	—	—	0.44	62.84
UAF 1st 12 Month Exposure Period	0°	—	33.7	0.52	0.74	—	—	0.39	64.64
	30°	—	33.04	0.43	0.57	—	—	0.4	65.55
	45°	—	36.95	—	0.27	—	—	0.42	62.37
POA 1st 12 Month Exposure Period	0°	—	31.13	0.74	1.3	—	—	0.36	66.47
	30°	—	31.71	0.3	0.57	—	—	0.35	67.07
	45°	—	27.25	0.68	0.66	—	—	0.56	70.86
UAA 1st 12 Month Exposure Period	0°	—	34.5	0.24	0.61	0.11	—	0.33	64.21
	30°	—	25.98	0.65	0.94	—	—	0.47	71.96
	45°	—	23.79	0.86	0.84	—	—	0.48	74.02

Calculated using the ISO⁽¹⁾ 25178 standard, the surface roughness parameter of the arithmetic mean height, S_a , is plotted for each test site for the first 12-month exposure period at each exposure angle in Figure 12.¹⁴ This parameter represents the average absolute deviation from the mean plane of the surface, giving a general indication of the overall roughness of the area scanned.

Figure 13 shows a 2D contour plot that presents the corrosion damage of the cleaned 1008 carbon steel coupons exposed over the first 12-month period at the PSCA test site. A pseudo-color view of each 10 mm x 10 mm area scanned is provided on the left and a corresponding black and white binarized photo is provided on the right for each exposure angle. Using the black and white binarized photo, the percentage of black to white pixels indicate possible areas of major corrosion damage. Coupons deployed at the 0° exposure angle show having a rougher surface than the coupons deployed at the 30° and 45° exposure angles. When observing the corrosion damage across all four test sites, the PSCA test site posted the highest percentages, however the coupons exposed at the 0° exposure angle do not all show higher percentages than the other two corresponding angles. It is reminded that only a 10 mm x 10 mm area was scanned on the skyward side of only one coupon from each test site for each exposure period at each exposure angle. More robust data could be obtained by scanning the whole coupon surface area of all coupons exposed. The trends found in the surface morphology data mirror that of the trends found in the corrosion rates calculated, shown previously in Figure 7.

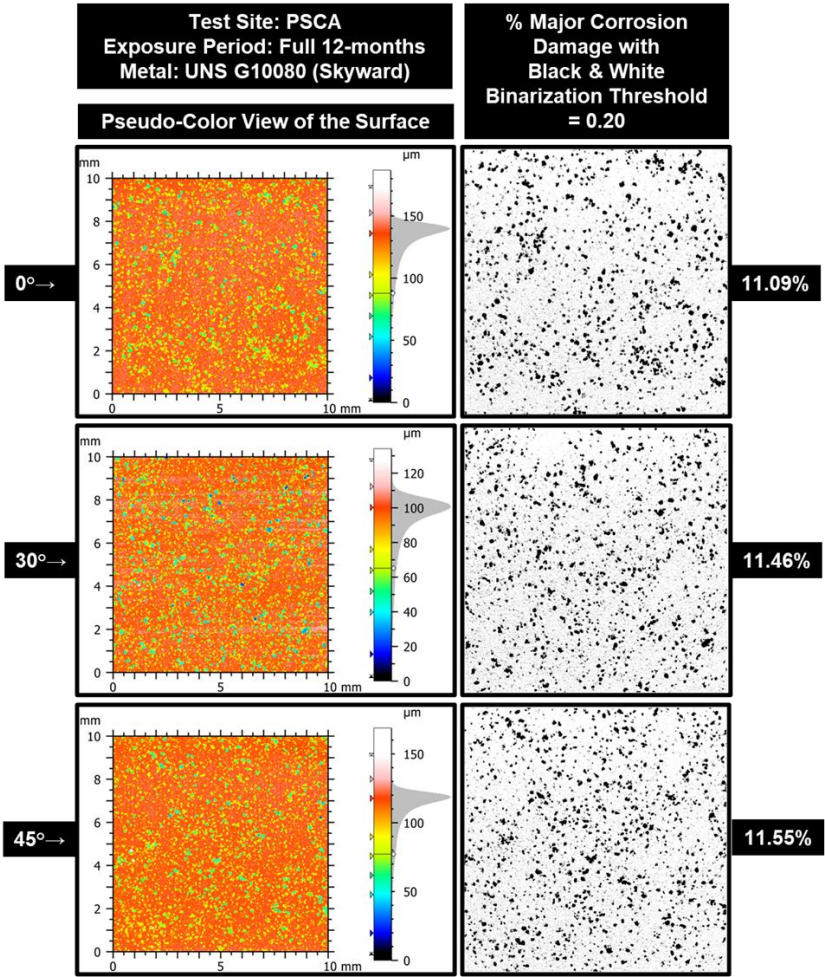


Figure 13: 2D Views and Percent Corrosion Damage of Cleaned 1008 Carbon Steel Coupons Exposed over the First 12-Month Exposure Period at the PSCA Test Site

CONCLUSIONS

Within the study of atmospheric corrosion, there exists a knowledge gap in the corrosivity of metals in colder regions and arctic environments. Where the complexity of corrosion increases, evaluation of the corrosivity of the atmosphere is made more difficult. With Alaska being a growing strategic location for commerce, military defense, refueling, and aerospace applications, further studies were conducted to enhance corrosion estimation in colder regions. Multi-angle corrosion test racks were deployed at four test sites across Alaska. Distinct in their environment, each were equipped with weather sensors and chloride candles. Analyses including weather data, mass loss data, along with SEM/EDXA and surface profilometry data were done to determine corrosion rates and compare their corrosivity classifications against well recognized standards. Of the four test sites, the PSCA test site resulted with the highest corrosion rate by a significant margin. This can be attributed to the aggressive weather parameters showing a high positive correlation between the time of wetness and corrosion rate and the high positive correlation between the measured chloride deposition rate and corrosion rate. Regarding exposure angle, there existed trends from all four test sites where coupons exposed at lower angles compared to the horizontal exhibited much less corrosion, outlining its significance. Using known corrosion rates through exposure of coupons and the estimation of corrosion rates through the dose-response function and the respective weather data, the categorization of corrosivity of the atmosphere of the test sites varied. Resulting, newer standardized models are required to increase accuracy in the estimation of corrosion in colder regions that include additional parameters such as the time of wetness and angle of exposure. The significance of these findings indicate that the studies of atmospheric corrosion in Alaska are a forefront for future atmospheric corrosion studies in colder regions alike.

FUTURE WORK

- Explore the correlation of corrosion rate to exposure conditions and weather parameters using additional data and determine the corrosivity category for the additional exposure periods and metal alloys.
- Create a dose-response function that represents the atmosphere of Alaska that incorporates the time of wetness and angle of exposure to estimate corrosion rates of metal alloys in colder regions.

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